

NORTHEAST OHIO, OH, USA

WMO: 725256

Lat: 41.778N

Lon: 80.696W

Elev: 924

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 04857

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	7.7	-6.8	3.9	4.3	-1.1	5.3	9.6	25.0	29.5	22.7	28.9	5.3	220	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	88.5	73.6	85.9	72.1	83.4	70.9	76.1	84.6	74.5	82.2	73.0	80.2	9.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	127.9	80.4	72.1	122.8	79.1	70.5	116.2	77.4	40.0	85.0	38.5	81.9	37.1	80.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	18.7	16.7	DB	-6.1	92.0	8.9	2.2	-12.5	93.6	-17.7	94.9	-22.7	96.1	-29.1	97.7	(4)
(5)			WB	-6.8	78.4	8.5	1.7	-13.0	79.7	-18.0	80.6	-22.7	81.6	-29.0	82.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	26.7	27.4	35.9	48.1	59.0	67.3	71.4	69.7	63.8	52.7	42.4	33.0		
	DBStd	18.22	11.87	11.60	11.49	9.79	8.97	6.80	5.41	5.26	7.19	8.43	8.97	9.15		
	HDD50	2854	725	635	460	152	22	0	0	0	1	67	262	530		
	HDD65	6134	1187	1052	902	514	231	52	9	17	108	392	679	991		
	CDD50	2821	3	3	24	94	300	519	664	610	416	151	33	4		
	CDD65	627	0	0	1	6	45	121	208	162	73	11	0	0		
	CDH74	5232	0	0	6	95	422	1071	1767	1232	565	72	2	0		
	CDH80	1436	0	0	0	18	91	332	524	305	156	10	0	0		
(14) Wind	WSAvg	7.2	9.3	8.6	8.2	8.1	6.5	5.9	5.0	4.7	5.3	7.1	8.3	9.0		
(15) Precipitation	PrecAvg	44.0	3.3	2.4	3.0	3.9	3.8	4.0	4.5	3.7	4.3	3.9	3.7	3.5		
	PrecMax	54.6	6.2	5.7	6.2	6.1	7.4	7.2	9.5	6.3	8.2	6.7	5.4	7.4		
	PrecMin	29.8	1.5	0.9	0.9	2.3	1.0	0.8	1.4	1.9	1.2	1.9	1.7	1.9		
	PrecStd	6.7	1.2	1.1	1.3	1.2	1.7	1.6	2.1	1.3	1.9	1.3	1.1	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	63.1	73.1	82.3	86.5	90.8	91.3	90.7	89.1	81.2	70.3	62.6		
		MCWB	55.1	53.5	59.5	64.1	70.6	73.7	74.7	74.9	73.6	69.8	58.5	55.9		
	2%	DB	54.7	54.4	65.2	76.2	82.4	87.8	88.4	86.3	84.3	75.1	64.1	55.4		
		MCWB	50.9	49.6	55.1	59.9	67.6	72.8	73.9	73.1	72.1	64.4	55.5	50.4		
	5%	DB	48.4	48.5	59.5	71.0	79.1	84.0	85.6	83.1	80.8	70.6	61.0	51.0		
		MCWB	44.5	43.4	51.7	57.2	66.0	70.7	72.7	71.6	69.7	62.6	53.6	47.4		
	10%	DB	43.2	43.1	54.4	65.5	74.8	80.6	82.7	80.9	76.4	66.4	56.7	46.0		
		MCWB	39.3	38.1	47.6	54.4	63.0	68.4	70.9	70.1	67.4	60.0	50.3	42.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.0	57.1	61.1	66.5	73.2	77.1	78.4	77.4	76.7	71.4	62.0	56.9		
		MCDB	59.7	61.3	71.6	78.3	83.3	87.2	87.5	86.8	85.0	79.5	67.8	60.2		
	2%	WB	51.3	50.0	57.1	62.2	70.1	74.4	76.2	75.3	73.7	67.7	57.4	51.8		
		MCDB	54.3	53.8	63.6	73.5	79.4	83.7	84.7	82.8	80.7	72.5	62.1	54.9		
	5%	WB	45.3	43.8	52.9	59.0	67.8	72.3	74.5	73.7	71.3	64.0	54.3	46.8		
		MCDB	48.3	48.3	58.4	67.8	76.4	81.2	82.0	80.2	77.9	68.7	59.7	49.9		
	10%	WB	39.7	38.3	47.9	55.7	65.5	70.3	72.9	72.0	69.0	60.5	50.9	42.5		
		MCDB	43.2	42.9	54.1	64.4	72.8	77.9	79.8	78.2	74.9	65.9	55.9	46.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	15.7	16.7	20.0	21.0	19.9	19.4	19.3	19.4	16.9	14.3	11.4		
		MCDBR	18.5	21.2	24.8	28.5	25.0	23.0	22.0	21.9	23.4	22.6	20.4	18.2		
		MCWBR	16.4	17.5	16.3	16.6	13.5	11.8	10.5	11.3	12.2	13.4	14.5	15.4		
	5% WB	MCDBR	18.0	20.7	22.3	25.1	21.8	20.1	18.2	18.0	20.3	19.5	18.2	17.2		
		MCWBR	17.0	18.3	16.2	16.5	12.5	11.4	9.6	10.2	11.2	13.2	14.5	15.8		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.334	0.349	0.383	0.419	0.443	0.443	0.428	0.384	0.362	0.319	0.300		
	taud		2.417	2.274	2.311	2.306	2.237	2.213	2.231	2.267	2.363	2.414	2.514	2.500		
	Ebn at Noon		269	274	284	281	272	265	264	264	269	262	262	261		
	Edh at Noon		25	34	36	40	43	45	43	41	35	29	23	22		
(44) All-Sky Solar Radiation	RadAvg		448	696	1035	1411	1701	1876	1882	1666	1328	839	534	335		
	RadStd		39	82	114	143	154	147	138	86	127	79	78	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	N/A

Nomenclature: See separate page